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Habitat selection and diet of the Capercaillie *Tetrao urogallus* in Abernethy Forest, Strathspey, Scotland

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Capsule Woodland characteristics are important for fully grown birds and for broods.

Aims To describe habitat selection and diet of the Capercaillie in ancient native pinewoods and Scots Pine *Pinus sylvestris* plantations, in order to provide a guide for forest management.

Methods Tree selection was studied in winter by locating droppings under trees along transects, and comparing those trees used by Capercaillie with those not used. Habitat selection by broods was studied by following radiotagged chicks or females with broods. Diet was described from the analysis of droppings.

Results In winter, Capercaillie preferred Scots Pines with wide crowns and large branches, and little surrounding Heather *Calluna vulgaris*, but avoided forest tracks. Female Capercaillie occurred in stands with a higher density of pines compared to males. The diet of fully grown birds comprised primarily Scots Pine needles between October and April. Scots Pine pollen cones were eaten during May, June and July. Bilberry *Vaccinium myrtillus* leaves and fruit were eaten in summer and autumn. Habitats used by broods had a higher percentage of *Vaccinium* sp. and smaller Scots Pines compared with random sites. Chicks hatch at the beginning of June, and in their first two weeks, their diet comprised Formicidae, Lepidoptera larvae and Coleoptera. In July and August, Lepidoptera larvae were rarely eaten but a higher percentage of chicks took vegetable material (particularly Bilberry fruit). In June, smaller numbers of invertebrates were generally trapped in mires of wooded bogs compared with areas of Heather and *Vaccinium* sp. in old woodland. Areas of *Vaccinium* sp. had the highest numbers of Formicidae, Coleoptera and Diptera, and especially Lepidoptera larvae. Numbers of Lepidoptera larvae declined during June and July as larvae pupated, explaining the reduction of Lepidoptera larvae in the diet of older chicks.

Conclusions Habitat management for Capercaillie should aim to provide old Scots Pine forest with a *Vaccinium*-rich field layer. Within Scotland, this may entail increasing the tree density in ancient native pinewoods but decreasing it in pine plantations. The removal or closure of tracks might increase the amount of woodland available to Capercaillie, by reducing human disturbance.

Within Britain, the Capercaillie *Tetrao urogallus* is restricted to the Highlands of Scotland, from Argyll in the south to Ross-shire in the north (Wilkinson *et al.* 2002). The current population is derived from introductions made in the early 19th century from Scandinavia, after extinction of the indigenous population (Ritchie 1920). The introduced birds increased and became established, but the population declined

during the last 30 years. Its range contracted by 64% between 1968–72 and 1988–91 (Gibbons *et al.* 1996), and counts estimated only 2200 birds during 1992–94 (Catt *et al.* 1998) and 1100 in 1998–99 (Wilkinson *et al.* 2002). As a result, the Capercaillie has been included in the red list of Birds of Conservation Concern in the United Kingdom and given high priority in the UK's response to the Biodiversity Convention (Anon 1995).

Highest densities of Capercaillie occur in ancient native pinewoods (Catt *et al.* 1998), which represent

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the remaining fragments of the Caledonian pine forest that once extended over much of the Highlands of Scotland (Steven & Carlisle 1959). These woods are recognized for their high levels of biodiversity, and so have an important conservation status (Anon 1995). Despite this importance, there has been only limited research on habitat use by Capercaillie in native pinewoods in Scotland (Picozzi *et al.* 1992).

Although the ancient native pinewoods have the highest densities of Capercaillie in Scotland, most of the population occurs at low densities in plantations (Catt *et al.* 1998). Given that silvicultural practices can be detrimental to Capercaillie habitat (Gjerde 1991b) and the Forestry Commission is supporting a scheme for the establishment of new native pinewoods (Forestry Commission 1989), there is a need to understand the habitat requirements of Capercaillie so that they can be provided for in existing and new plantations. There has been a number of studies of the habitats of Capercaillie, describing patterns in different seasons or during events of the annual cycle: autumn (Rolstad 1988a), summer (Rolstad *et al.* 1988), lekking (Picozzi *et al.* 1992), nesting (Jones 1984) and attending broods (Storch 1994). Often, however, habitat descriptions have not been quantified or conclusions have been general, indicating, for example, that Capercaillie prefer old pine-dominated forest (Rolstad & Wegge 1989). Ideally, more detailed descriptions and measurements are required in order to recommend management prescriptions. For example, Picozzi *et al.* (1992) provide a simple yet quantitative measure for stand types.

Here, the habitat choice of fully grown Capercaillie in winter and brood habitats in summer were examined. The work was carried out at Abernethy Forest, which is the largest remaining area of ancient native pinewood in Scotland (2452 ha) (Forestry Commission 1998) but also contains plantations (c. 1000 ha).

An understanding of the habitat requirements should also include knowledge of diet. There have been a number of studies of Capercaillie diet, usually restricted to one part of the year (Zwickel 1966, Pulliainen 1970, Borchtschevski 1995), and few throughout the year (Lieser 1996, Storch *et al.* 1991). Therefore, the seasonal changes in diets of fully grown birds and chicks were described in this study. In order to explore the relationship between chick diet and habitat preferences, we sampled invertebrates in different habitats and at different times in the summer when Capercaillie chicks are feeding on invertebrates.

STUDY AREA AND METHODS

The wooded part of Abernethy Forest RSPB Reserve (57°15'N, 3°40'W) extends over about 30 km² of the northern slopes of the Cairngorm Mountains and extends down into Strathspey. The history of the forest has been described by Steven & Carlisle (1959) and O'Sullivan (1973), and the current woodland structure by Summers *et al.* (1997). Almost all the conifers are Scots Pines *Pinus sylvestris*.

Habitat selection in winter

Twenty transects of about 2 km long, U-shaped and running north–south were set randomly throughout the forest. Eighty Scots Pines were chosen at 25-m intervals along each transect and marked. In order to determine which trees the Capercaillie used, droppings were recorded under the crowns of the marked trees in February/March 1991–94.

The following measurements were taken from the marked trees along the transects: tree height (using a clinometer at a measured distance from the tree); diameter at breast height (DBH); percentage of the vertical height of the trunk that had foliage; crown diameter; presence of branches of diameter c. < 5, 5–15 and > 15 cm less than halfway up the trunk; distance to nearest neighbour and density of trees greater than 1 m tall within a 4–25 m radius of the tree (the different radii depended on the density of trees). The stand structure in terms of two principal components scores (PRIN1 score and PRIN2 score) (Picozzi *et al.* 1992) was described. As trees become broader, have a deeper crown and fewer interlocking branches with neighbours, their PRIN1 score increases. Such trees are typical of ancient native pinewood in Scotland. With increasing PRIN2 scores, trees are closer together, more even-aged and their branches interlock. This is typical of trees in plantations (see Picozzi *et al.* 1992 for details). The percentages of the field layer covered by Heather *Calluna vulgaris* and *Vaccinium* sp. (Bilberry *V. myrtillus* and Cowberry *V. vitis-idaea*) were visually estimated within a 10-m radius of each tree. Cowberry forms only 4–25% of the *Vaccinium* sp. in different stand types (Parlane *et al.* unpubl. data), so the majority of this vegetation type refers to Bilberry. Distances to the nearest track or road were measured from 1:25 000 maps after the transects had been plotted using a global positioning system.

Characteristics of trees that were used by the birds were compared with those for trees not used along the

transects using a logistic model (Aitkin *et al.* 1989, Crawley 1993). The y variable was set at 1 for Capercaillie trees and 0 for non-Capercaillie trees, and a logit link function applied. Model building was in a step-up procedure, with each variable tested in turn against the response variable. The variable resulting in the greatest significant reduction in deviance (tested as chi-squared) was selected at each stage. After the addition of a new variable, the remaining variables were again tested in turn until a minimum adequate model was achieved. However, the vegetation variables were entered into the model before testing for the effect of distance to tracks.

Additional data were obtained along the 40 km of transects at trees from which Capercaillie were flushed. In this way, sexual differences in tree use could be assessed. The transects were walked each month from December to March from 1991/92 to 1998/99. Variables measured at these trees included: tree height; crown percentage and diameter; DBH; nearest neighbour distance and tree density (see above).

Diet of fully grown birds

Twenty Capercaillie droppings from fully grown birds were collected arbitrarily each month within *Pinus sylvestris*–*Hylocomium splendens* woodland (W18) of the National Vegetation Classification (Rodwell 1991), although bog habitat (*Erica tetralix*–*Sphagnum papillosum* (M18)) was also visited in April. When females were incubating, their very large droppings produced during breaks in incubation were collected. In order to quantify the proportions of the food items in the droppings, a subsample was taken from each dropping and spread with water in a gridded petri dish, and observed under a binocular microscope. At each of 100 intersections of the grid, the nearest food fragment was identified. Frequencies of each food type were averaged for all droppings collected in a given month.

Diet of chicks and habitats of broods

Capercaillie broods were located in mid-June and again in late July/early August using trained dogs. Captured chicks were held in cotton bags until they defecated. Droppings were stored in 70% alcohol. Later, droppings were teased apart under a binocular microscope and the fragments of invertebrates and plants identified from a reference collection or by specialists. Diet was described in terms of percentage occurrence of each category, rather than by a volumetric or numerical

method because differential digestion of the different food groups would bias the results (Picozzi *et al.* 1999). Chicks were aged from wing-length (Spidsø & Stuen 1991).

Small chicks (less than 100 g) were radiotagged with 1-g transmitters glued to the back with superglue, and large chicks (over 600 g) with 18-g necklace tags. Attempts were made to relocate broods daily. Details of the habitat at brood sites were recorded as follows: the nearest ten trees around the brood site were measured (height and DBH) and medians calculated; tree density recorded as on the transects; the percentage top cover of Heather, *Vaccinium* sp. and other plants was based on the presence/absence of these plants at the 50 intersections of a 10 × 5 m grid, with the brood site positioned centrally; ten recordings were made of the heights of *Vaccinium* sp. using a sliding polystyrene disc (diameter 24 cm and mass 30 g) on a graduated stick and medians calculated; distances to forest edge and nearest track were measured from maps. The locations where large chicks were captured were not described because of the risk of the brood moving ahead of us during the initial search and not being in its chosen habitat once located. A random site was also described at 100 m on a random compass bearing from each brood site. A comparison between brood and random sites was made in the same way as the comparison between Capercaillie and non-Capercaillie trees on the transects. However, as the habitat data collected daily from a given brood were not independent, medians were calculated for each brood prior to analysis. In order to give more weight to the data from broods with a greater number of observations, the square root of the sample size for each brood was used as the weight. Again, vegetation variables were tested before distances to track and forest edge.

Invertebrate sampling

Four sites were selected for invertebrate sampling: Rynettin (grid ref. NJ017157), Bognacruie (NJ034154), Dell Road (NJ010161) and the Ice Wood (NJ025135). At each site, four contrasting vegetation types were sampled: Heather, *Vaccinium*-flats (growing on flat ground), *Vaccinium*-hummocks (growing on hummocky ground) and mire vegetation composed mainly of Heather, Bog Cotton *Eriophorum* sp., *Erica tetralix* and grasses. The Heather and *Vaccinium* types fell within *Pinus sylvestris*–*Hylocomium splendens* woodland (W18), whilst the mire vegetation in the wooded bogs fell within the *Erica tetralix*–*Sphagnum papillosum* (M18) community (Rodwell 1991).

Sampling of invertebrates was carried out using pit-fall traps and a sweep-net in 1991 and 1992. Four pit-fall traps were set in a line approximately 4-m apart in each habitat at each site. All pit-fall traps were installed on 1 June. The traps were emptied on 15 June and reset. They were then emptied and removed on 1 July. Invertebrates were sampled in early and late June using a sweep-net (cotton bag on metal rim with 40 cm diameter) when the vegetation was dry along four lines, 4-m apart at each site, and involved 25 sweeps in each vegetation type at each site.

An additional programme of sweep-netting was carried out on *Vaccinium*-flats in 1994 to examine trends in invertebrate numbers in June and July. Trapping was carried out at the above four sites (six samples per site) in mid-June (17 June), early July (1 July), mid-July (15 July) and late July (30 July).

Invertebrates were removed from the traps, sorted and counted and then stored in 70% alcohol and later identified using Chinery (1977) and specialists. Lepidoptera larvae were frozen so they would not lose their colour, which is important in identification.

The analysis of the trapping data was split according to trap type and performed only on those groups where a large sample was caught. For each invertebrate group, Poisson regression models were fitted using a logarithmic link function (Aitkin *et al.* 1989). Site effect was introduced first in all models followed by the main effects that we were particularly interested in; vegetation type, early/late trapping period and year. Overdispersion was accounted for using Pearson's χ^2 (Crawley 1993).

Vegetation height of the field layer was measured using a sliding polystyrene disc (24 cm in diameter and mass of 30 g) on a graduated stick placed at 1-m intervals. Twenty measurements down the line of the pit-fall traps were taken at each site.

RESULTS

Habitat selection in winter

Scots Pines used by Capercaillie tended to be large and at a low density (Table 1). Thus, the associated stands tended to have higher PRIN1 and PRIN2 woodland structure scores compared with those not used (Fig. 1, Table 1). The associated field layer tended to have less Heather and more *Vaccinium* sp. than under trees not used by Capercaillie. Notably, trees used by Capercaillie also tended to be further from tracks or roads.

In the logistic model, transect number was introduced first to account for spatial differences. Thereafter, the significant explanatory variables were crown breadth, Heather cover and the presence of large branches low down on the tree (Table 2). Capercaillie preferred Scots Pines with wide crowns, large branches and where the surrounding field layer had a low percentage cover of Heather. There was a positive correlation between wide crowns and DBH ($r = 0.93$, $P < 0.001$) and height ($r = 0.68$, $P < 0.001$) suggesting a preference for large trees. After these woodland structure characteristics had been entered into the model, the distance to the nearest track was found

Table 1. Median values and interquartile ranges (IQR) for Scots Pines *Pinus sylvestris* used and not used by Capercaillie *Tetrao urogallus* in winter at Abernethy Forest. The surrounding field layer was also described. Sample sizes = 133 for Capercaillie trees and 1294 for non-Capercaillie trees.

Parameter	Used by Capercaillie		Not used by Capercaillie		Mann-Whitney <i>U</i>	<i>P</i>
	Median	IQR	Median	IQR		
Height (m)	13.9	10.1–16.8	9.0	4–14.5	51 590	< 0.001
DBH (cm)	47.8	29.3–62.7	21.4	8–38.2	37 751	< 0.001
Crown height (% of tree)	70	60–80	70	50–90	88 322	0.442
Crown diameter (m)	8	6–12	4	2–6.5	34 862	< 0.001
Nearest neighbour (m)	4.5	2.5–7.5	2.5	1.5–4.5	57 347	< 0.001
Density (no./ha)	200	110–430	470	220–1150	118 010	< 0.001
Heather (%)	30	20–45	45	20–80	110 467	< 0.001
<i>Vaccinium</i> sp. (%)	35	20–50	15	5–35	50 868	< 0.001
PRIN1 score	6.0	3.5–8.5	3.5	1.0–6.0	48 765	< 0.001
PRIN2 score	3	2–3	2	1–4	65 858	< 0.001
Distance (m) to						
Track or road	175	100–250	125	50–225	70 202.5	< 0.001
Track	175	100–250	150	50–250	71 020.5	0.001

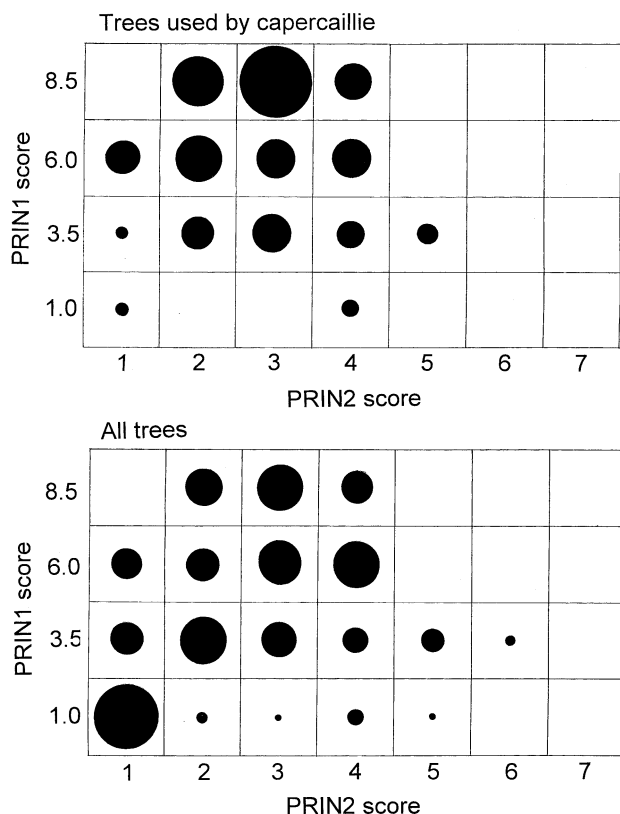


Figure 1. The percentages of records of Capercaillie in different stand structures at Abernethy Forest, based on PRIN1 and PRIN2 scores (Picozzi *et al.* 1992). The percentages available are also shown. The areas of the dots are proportional to the percentages.

to provide additional explanatory value (Table 2). Birds were more likely to occur in trees further from tracks.

An assessment was made of the trees used by male and female Capercaillie in winter when they were flushed on the transects. There were no significant differences between the trees used by males and females for most of the variables measured, with the exception of tree density (Table 3). Females used stands with a higher tree density compared to males.

Diet of fully grown birds

The diet of fully grown Capercaillie consisted largely of Scots Pine needles, which were eaten throughout the year. This varied between 13.5% in droppings in August to over 95% during November to April (Table 4). Four droppings collected in bog habitat in April were almost entirely (97%) composed of flower buds of *Eriophorum* sp. During May, June and July, pollen cones of Scots Pines became the main food of Capercaillie, including incubating females. Pollen cones were well formed by mid-May and shed their pollen in late May and early June. At the end of June/early July, the small cones were shed from their stalks. Other food that was important in summer included Bilberry leaves and *Carex* seeds. During July to September, Bilberry berries were eaten (Table 4).

Habitat use by broods

Data were obtained from 14 broods at 55 sites in June, and 12 broods at 115 sites in late July/August. Habitat data for each brood were averaged prior to analysis. In an initial comparison of brood and random sites, there was a tendency for broods to avoid areas with Heather and to prefer areas with *Vaccinium* sp. (Table 5). There was no significant difference in the heights of *Vaccinium* sp. between sites used by broods and random sites. Not all brood sites were within woodland; records were also made of broods in areas of moorland and bogs surrounded by pinewood.

A logistic regression analysis showed that the most significant variable affecting habitat choice for broods was the amount of *Vaccinium* sp., and Scots Pine height had an additional effect. The equation had the form: $\log_e(P/(1 - P)) = 2.217(\pm 1.118 \text{ se}) + 0.054 (\pm 0.011 \text{ se}) \% \text{ Vaccinium} - 0.296 (\pm 0.086 \text{ se}) \text{ tree height}$, where P is the probability that a capercaillie brood occurs within a given habitat. In order to test whether there was an optimum percentage cover of *Vaccinium* sp., the

Table 2. Estimates of the coefficients in a logistic model predicting the probability of Scots Pines being used by Capercaillie in winter at Abernethy Forest. The variables tested in the model included those in Table 1. The percentage of the deviance accounted for was 30.6%.

Explanatory variable	Coefficient	se	t	P
Intercept	-4.870	± 1.054	-4.62	< 0.001
Crown breadth	0.2285	± 0.0418	5.47	< 0.001
Percentage of Heather	-0.0242	± 0.0056	-4.32	< 0.001
Branch thickness:				
5–15 cm	0.7092	± 0.2988	2.37	< 0.05
> 15 cm	0.2796	± 0.4413	0.63	> 0.1
Distance to nearest track	0.00548	± 0.00116	4.72	< 0.001

Table 3. Median values and interquartile ranges (IQR) for Scots Pine trees used by male and female Capercaillie in winter at Abernethy Forest. Sample sizes: 64 for males and 55 for females.

Parameter	Used by male Capercaillie		Used by female Capercaillie		Mann–Whitney <i>U</i>	<i>P</i>
	Median	IQR	Median	IQR		
Height (m)	15.1	13.2–16.4	16.0	12.5–18.2	1586.5	0.355
DBH (cm)	50.2	37.9–60.0	43.6	39.3–53.7	2056.5	0.114
Crown height (% of tree)	50	40–60	50	40–65	1630.5	0.484
Crown diameter (m)	6.5	5.0–8.6	6.0	4.5–8.0	1893.5	0.201
Nearest neighbour (m)	4.4	2.8–7.0	3.7	2.9–6.1	1933.5	0.355
Density (no./ha)	195	130–350	350	200–565	1184.5	0.002

Table 4. The diet of fully grown Capercaillie and chicks at Abernethy Forest. The data for fully grown birds are expressed as the percentage of items found in droppings each month. The data for chicks refer to the percentage occurrence of diet items in droppings.

	Fully grown birds												Incubating females	Chicks	
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	May/June	June	July/Aug
Scots Pine needles	99.6	99.7	98.8	95.0	33.1	22.0	26.4	13.5	68.5	72.2	100.0	96.8	13.3	0.0	0.0
Scots Pine pollen cones	0.0	0.0	0.0	0.0	47.0	44.0	30.0	0.0	0.0	0.0	0.0	0.0	76.4	0.0	0.0
Bilberry leaves	0.0	0.0	0.0	0.0	12.2	27.4	3.2	7.1	20.0	8.6	0.0	2.7	8.0	15.4	3.7
Bilberry fruits	0.0	0.0	0.0	0.0	0.0	0.0	10.3	65.3	7.4	0.6	0.0	0.5	0.0	26.9	74.1
Cowberry leaves	0.2	0.0	0.0	0.0	1.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Cowberry fruits	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Heather leaves	0.2	0.3	1.2	0.2	1.4	0.4	5.0	1.8	2.7	18.1	0.0	0.0	1.1	0.0	18.5
<i>Eriophorum</i> sp. flowers	0.0	0.0	0.0	4.8	0.0	6.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
<i>Carex</i> sp. seeds	0.0	0.0	0.0	0.0	0.0	0.0	24.3	11.2	0.7	0.1	0.0	0.0	1.0	0.0	29.6
Moss	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	11.1
Grass	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	7.7	3.7
Unidentified vegetation	0.0	0.0	0.0	0.0	5.0	0.0	0.6	0.3	0.1	0.3	0.0	0.0	0.1	7.7	18.5
Arachnids	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	30.8	7.4
Lepidoptera larvae	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	73.1	7.4
Coleoptera adults	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	73.1	29.6
Coleoptera larvae	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	3.9	0.0
Diptera adults	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	19.2	3.7
Formicidae adults	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	50.0	70.4
Unidentified invertebrates	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.6	0.1	0.0	0.0	0.1	0.0	0.0
No. of droppings	20	20	19	20	20	20	13	20	20	20	20	20	21	26	27

quadratic term was fitted but found to be not significant ($\chi^2 = 0.07$, ns) indicating that the simple model of increasing cover of *Vaccinium* sp. best explained the probability of sites being used by Capercaillie broods. The age of the brood (early versus late broods) had no significant effect on the model.

Diet of chicks

A high proportion of chicks in June, aged between 2 and 13 days ate Formicidae (*Formica* sp. and *Myrmica* sp.), Coleoptera and Lepidoptera larvae (Table 4). Only a few were eating plant material. By the time the chicks were 6–8 weeks old, most were eating plant material, particularly Bilberry fruit, but also sedge

Carex sp. seeds, ericaceous leaves and moss (Table 4). The only insects to occur frequently were Formicidae and Coleoptera.

Invertebrate abundance in different habitats

Heather formed the tallest vegetation, and the *Vaccinium* sp. was intermediate in height (Table 6).

Fewest invertebrates were trapped in the mires compared with other vegetation types, except for ground-living Arachnids (Table 7). Heather had the most Arachnids and Coleoptera living in the field layer. Formicidae were most abundant on *Vaccinium*-flats. There were consistent trends within the two *Vaccinium* habitats for Coleoptera (in pit-falls), Diptera

Table 5. Median values and interquartile ranges (IQR) for habitat variables at locations of 26 Capercaillie broods and at random locations in Abernethy Forest.

Parameter	Brood sites		Random sites		Mann–Whitney <i>U</i>	<i>P</i>
	Median	IQR	Median	IQR		
Tree height (m)	13.8	11.6–16.5	15.7	12.8–16.8	413	0.170
Tree DBH (cm)	38	18–45	35	26–43	332.5	0.920
Tree density (no./ha)	305	170–650	320	150–580	309	0.595
Heather cover (%)	35	14–52	56	22–86	446.5	0.047
<i>Vaccinium</i> cover (%)	47	29–68	34	4–55	208	0.017
<i>Vaccinium</i> height (cm)	24	21–27	23	21–26	263.5	0.641
Distance (m) to						
Forest edge	610	275–925	540	200–875	326.5	0.833
Nearest track	150	100–325	150	75–250	287	0.350

Table 6. Mean heights (cm $\pm$ sd) of vegetation where invertebrate sampling was carried out.

Site	Heather	<i>Vaccinium</i> - hummocks	<i>Vaccinium</i> - flats	Mire vegetation
Rynettin	36 $\pm$ 10	24 $\pm$ 11	24 $\pm$ 10	16 $\pm$ 5
Bognacruie	47 $\pm$ 9	24 $\pm$ 9	20 $\pm$ 10	15 $\pm$ 7
Ice wood	35 $\pm$ 10	22 $\pm$ 10	16 $\pm$ 5	16 $\pm$ 4
Dell Road	47 $\pm$ 16	23 $\pm$ 6	23 $\pm$ 7	24 $\pm$ 15

and Lepidoptera larvae, in that they generally had larger numbers than either Heather or the mires. By far the biggest difference between vegetation types was for Lepidoptera larvae, which were 10.3-times and 12.6-times more abundant on *Vaccinium*-hummocks and *Vaccinium*-flats, respectively, than on Heather (Table 7). *Hydriomena furcata* and *Operophtera brumata* comprised most (95.8%) of the Lepidoptera larvae in

1991 but only 22.8% in 1992. Most Lepidoptera larvae in 1992 were micro-moths.

There were more Arachnids (in pit-falls), Formicidae and Coleoptera (in pit-falls) caught in 1992 compared with 1991, but there were fewer Arachnids (in sweep-nets), Coleoptera (in sweep-nets) and Diptera (Table 7).

Comparisons between early and late June showed that there were fewer Arachnids (in pit-falls) Formicidae, Coleoptera and Lepidoptera larvae in late June compared with early June. Diptera were more abundant in late June (Table 7). In 1994, there was an extended examination of trends in abundance during the summer, just within *Vaccinium* sp. Between mid-June and late July, Lepidoptera larvae ($\chi^2_3 = 326.5$, $P < 0.001$) and Diptera ($\chi^2_3 = 117.7$, $P < 0.001$) declined markedly (Fig. 2). There were no significant changes in the abundance of Coleoptera ($\chi^2_3 = 2.1$, ns), Formicidae ($\chi^2_3 = 6.8$, ns) and Hemiptera ($\chi^2_3 = 6.6$, ns).

Table 7. Indices of abundance (95% confidence limits in brackets) derived from Poisson regression models fitted to numbers of invertebrates trapped by pit-falls (PF) and sweep-netting (SN) in four vegetation types, during two periods in June and two years. The site effect and overdispersion were taken into account in all the models. All values are related to numbers trapped in Heather in early June 1991 when the index was set at 1.

	Arachnids		Formicidae PF	Coleoptera		Diptera		Lepidoptera larvae SN
	PF	SN		PF	SN	PF	SN	
Mire	2.9 (2.1–4.0)	0.3 (0.2–0.4)	0.5 (0.2–1.1)	0.6 (0.4–0.9)	0.2 (0.1–0.4)	1.0 (0.6–1.8)	0.9 (0.6–1.5)	0.3 (0.06–1.5)
<i>Vaccinium</i> - flats	1.3 (0.9–1.9)	0.6 (0.4–0.8)	2.3 (1.3–4.2)	2.0 (1.5–2.6)	0.2 (0.1–0.3)	2.4 (1.4–3.9)	1.8 (1.2–2.7)	12.6 (5.7–28.2)
<i>Vaccinium</i> - hummocks	0.6 (0.4–1.0)	0.6 (0.4–0.8)	0.9 (0.5–1.9)	2.1 (1.6–2.7)	0.1 (0.07–0.2)	5.8 (3.6–9.1)	1.3 (0.8–2.0)	10.3 (4.6–23.2)
1992	1.6 (1.3–2.0)	0.7 (0.6–0.9)	2.3 (1.4–3.9)	3.0 (2.4–3.6)	0.1 (0.09–0.2)	0.6 (0.5–0.8)	0.6 (0.4–0.8)	–
Late June	0.5 (0.4–0.6)	1.6 (1.2–2.1)	0.2 (0.08–0.3)	0.6 (0.5–0.7)	–	1.7 (1.3–2.2)	–	0.2 (0.1–0.3)

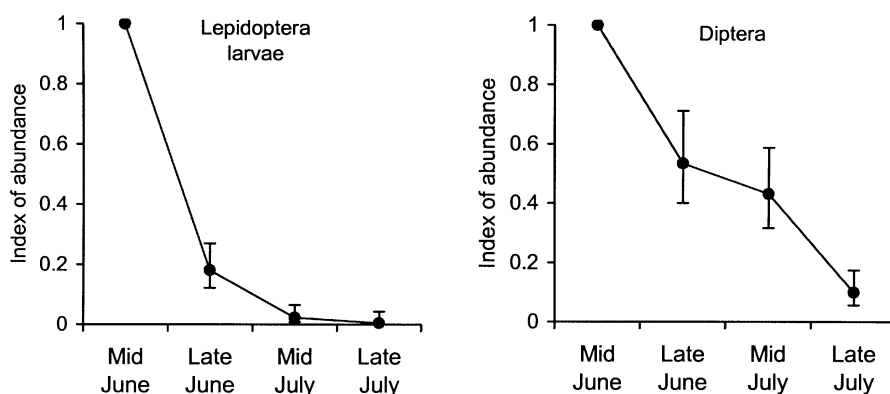


Figure 2. Changes in the indices of abundance of Lepidoptera larvae and Diptera in *Vaccinium* sp. during June and July 1994 at Abernethy Forest. Vertical lines show 95% confidence limits. The index in mid-June was set at 1 for each group.

DISCUSSION

Habitat selection and diet

Throughout its range, the Capercaillie is found in a variety of conifer and mixed conifer/broadleaf woodlands, comprising oak *Quercus* sp., Beech *Fagus sylvatica*, spruce *Picea* sp. and Scots Pine (Dement'ev *et al.* 1951). However, the association with Scots Pine woodland is strongest (Seiskari 1962). As with other studies (Zwickel 1966, Pulliainen 1970), fully grown Capercaillie at Abernethy Forest fed mainly on Scots Pine needles during winter (Table 4), although they have been reported to feed on the needles of other conifer species, such as Norway Spruce *Picea abies*, Silver Fir *Abies alba* and Sitka Spruce *Picea sitchensis* (Lieser 1996, Storch *et al.* 1991, Picozzi *et al.* 1996).

In a review of habitat use, Rolstad & Wegge (1989) noted that Capercaillie are usually associated with forests older than 70 years. The association with old forest can be explained in terms of basic needs for food and movement. We found that Capercaillie preferred trees with a broad crown and large branches (Table 2). These trees provide firm perches and enable the birds to move about the crown when feeding. Males weigh c. 4 kg (Cramp & Simmons 1980) so stout branches are beneficial. In addition, the food value of needles from older pines is perhaps better than from young trees because of the higher energy content (Lindén 1984).

Gjerde (1991a) found that there was an optimum conifer density for Capercaillie. Both sexes avoided forests with densities of less than 500 trees/ha and males avoided areas with densities exceeding 1000 trees/ha. In a tree-felling experiment, where the forest was thinned to 300–400 trees/ha, the area was abandoned by Capercaillie (Gjerde 1991b). In contrast, the average tree density in areas used by Capercaillie in winter

at Abernethy Forest was only 200 trees/ha (Table 1). Even the average density used by the females (350/ha) was low compared with the values given by Gjerde (1991b). It is possible that the optimum structure does not exist at Abernethy Forest, given its history of human use and grazing by deer and Sheep *Ovis aries* (O'Sullivan 1973), and perhaps the low density of old trees approximates to this optimum. Alternatively, it is possible that features other than tree density are important in habitat selection.

At Abernethy Forest, pollen cones of the Scots Pines were eaten in summer by both male and female Capercaillie. Pulliainen & Tunkkari (1991) have noted the importance of a high nitrogen and phosphorus diet (male catkins of Downy Birch *Betula pubescens*, flower buds of *Eriophorum vaginatum* and stems and tips of horsetails *Equisetum* sp.) prior to egg-laying in Capercaillie, but the pollen cones develop too late for this. However, a diet of pollen cones may aid recovery from egg-laying (O'Connor 1984). In addition, both sexes may benefit from this protein-rich food when moulting (Harrison & Evans 1985).

The review by Rolstad & Wegge (1989), a study by Storch (1993) and our own work showed that Capercaillie prefer woodland with Bilberry. Bilberry is deciduous and is important to Capercaillie in summer and autumn – fully grown Capercaillie eat the leaves and berries (Table 4) and the chicks eat the invertebrates associated with the plant (Kastdalen & Wegge 1985, Storch 1994, Picozzi *et al.* 1999, Table 4). In our study, the average cover of *Vaccinium* sp. at brood sites was 47%, and the average height was 24 cm.

Invertebrate food for chicks

Capercaillie broods selected areas with *Vaccinium* sp.

and these areas contained larger numbers of ground-living Coleoptera, Formicidae and Lepidoptera larvae than other vegetation types (Table 7). Bilberry leaves have a higher nitrogen content than Heather (Kerslake *et al.* 1996) and therefore provide a better food for Lepidoptera larvae, which in turn are food for predatory Coleoptera and Formicidae. All these invertebrates are food sources for Capercaillie chicks (Kastdalen & Wegge 1985, Spidsø & Stuen 1988, Picozzi *et al.* 1999). Capercaillie chicks also eat the berries of Bilberry when they are older. Thus, the rich food supplies found in Bilberry might account for the Capercaillie's preference for rearing broods in this habitat.

The decline of Lepidoptera larvae in the diet of chicks between June and July/August (Table 4) is probably explained by the declines in availability of the main species, *Hydriomena furcata* and *Operophtera brumata*, which feed primarily on Bilberry between April and June, and then pupate (Skinner 1984). The time of pupation may be related to the decline in nitrogen and increase in toughness of *Vaccinium* sp. leaves as the summer progresses (Kerslake *et al.* 1996).

Management implications for Scottish pinewoods to benefit Capercaillie

There are two contrasting types of conifer forest in Scotland; ancient native pinewood composed mainly of Scots Pine and birch, and plantations composed of single or mixed conifer species, including non-native conifers (Summers *et al.* 1999). The two types of woodland require different management prescriptions for Capercaillie to achieve old Scots Pine woodland with Bilberry. The ancient native pinewoods were selectively felled and heavily browsed by deer and Sheep in the past, leading to a park-like appearance of scattered trees at many sites (Steven & Carlisle 1959). Heather dominates the field layer but Bilberry is also common (Summers *et al.* 1999). Thus, a reduction in Sheep and deer densities might allow tree regeneration to occur, and with the increased shading of the field layer, Heather cover would decline and Bilberry increase (Parlane *et al.* unpubl. data), thereby benefiting Capercaillie. A reduction in herbivores would also reduce direct browsing on Bilberry (Baines *et al.* 1994).

In contrast, plantations in the Highlands are composed of differing proportions of Scots Pine, Lodgepole Pine *Pinus contorta*, Sitka Spruce, Norway Spruce and larches *Larix* sp. (Summers *et al.* 1999). Although Capercaillie will live in spruce forests and eat their

needles (Picozzi *et al.* 1996), Scots Pine woodland is preferred (Wilkinson *et al.* 2002). Therefore, to benefit Capercaillie, Scots Pines should be planted in new plantations or used when restocking a site. Plantations, particularly in their early stages, have tree densities that appear to be too high for Capercaillie (Gjerde 1991a) partly because the field layer is suppressed at canopy closure. When plantations become older and are thinned, more light reaches the ground, allowing the development of Bilberry. Thus, with appropriate thinning (Parlane *et al.* unpubl. data) and by increasing the rotation length so that the stands remain at the older stages when Bilberry is abundant for longer, plantations can be improved for Capercaillie (Sjöberg 1996).

Some observers have noted the importance of mixed-species forests for Capercaillie. Helle *et al.* (1990) found that spruce and birch in a pine forest increased the preference for a wood by Capercaillie. Although Capercaillie may benefit from some spruces in a forest, this is not seen as an option for the management in ancient native pinewoods where the aim is to have only native species (Anon 1995, Taylor 2000). There are no native spruces in Scotland. However, this might be an option for commercial pine plantations. Broad-leaved trees were once more common in the ancient native pinewoods of eastern Scotland but were selectively cut for firewood and charcoal or regarded as 'weeds' (Steven & Carlisle 1959, Summers *et al.* 1999). It would seem that the introduction of broadleaved trees, particularly Aspen *Populus tremula* (Rolstad 1988b) and birch, might benefit Capercaillie by providing spring food (Pulliainen & Tunkkari 1991).

Tracks, as well as other sandy sites (river banks and bases of upturned trees), are used by Capercaillie for collecting grit and for dust-bathing. Despite these attractive attributes of tracks, there was an avoidance of woodland close to tracks by fully grown birds in winter (Table 2), suggesting that human disturbance may be displacing Capercaillie and reducing the amount of woodland that can be fully occupied. In this case, removal or closure of tracks might benefit Capercaillie.

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